

# Lecture 2: Input Devices

**Keyboard · Mouse · Touch Screens · Joystick · Stylus · Scanners (OCR, OMR, MICR, Bar Code) · Biometric Devices · Digital Cameras**

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## Learning Outcomes

- Recall the Input Unit from earlier lectures, and explain why individual input devices are studied in mechanism-level detail.
- Explain, step by step, how a keyboard and a mouse each convert a physical action into computer input.
- Differentiate between resistive and capacitive touch screens, explaining how each detects a touch.
- Explain how a joystick and a stylus each work, including the difference between a passive and an active stylus.
- Explain, step by step, how OCR, OMR, MICR, and bar code readers each work, and identify a typical use for each.
- Explain what a biometric device is, and describe, step by step, how fingerprint recognition works.
- Explain, step by step, how a digital camera captures an image.
- Answer short, medium, and long-answer (up to 15 marks) examination questions on this topic with full step-by-step explanations, diagrams, and examples.

## 1 Recap — From Storage to How Data Gets In

Lecture 1 studied how a computer remembers data, once that data is already inside the system — memory, cache, and secondary storage. This lecture goes one step further back: how data gets INTO the computer in the first place. Every input device covered here does the same fundamental job — converting some real-world physical action or object into binary data — but each one does it through a completely different mechanism.

### Important Point for Exams

As with Lecture 1, this topic rewards mechanism-level understanding. Examiners rarely just ask you to name a device; they ask you to explain HOW it works, step by step, or to differentiate between two devices that do a similar job in different ways (such as OMR vs MICR, or a passive vs active stylus).

## 2 What Is an Input Device?

**An input device** is any hardware component that allows data, instructions, or physical actions from the outside world — usually from a human being — to be converted into binary form and supplied to the computer for processing. This lecture studies nine categories of input device in detail, grouped by the kind of real-world action or object each one converts.

## 3 The Keyboard

The keyboard is the primary input device used to enter text, numbers, and commands into a computer, arranged in the widely used QWERTY layout (named after the first six letters on the top row of letters).

A standard keyboard contains different types of keys according to their functions.

**The main groups of keys are:**

1. **Alphanumeric Keys:** These include alphabet keys **A–Z** and number keys **0–9**. They are mainly used for typing text and numbers.
2. **Function Keys:** These are **F1 to F12**, located at the top of the keyboard. They are used to perform specific functions. For example, **F1** is commonly used to open Help.
3. **Control Keys:** Keys such as **Ctrl, Alt, Shift, Esc, and Windows key** are control keys. They are often used alone or in combination with other keys to perform different operations.
4. **Navigation Keys:** These include **Arrow keys, Home, End, Page Up, and Page Down**. They are used to move the cursor and navigate through documents.
5. **Special/Editing Keys:** Keys such as **Enter, Spacebar, Backspace, Delete, Tab, and Caps Lock** are used for typing and editing operations.
6. **Numeric Keypad:** It is generally located on the right side of a full-sized keyboard. It contains **0–9** and mathematical operators such as **\*, +, -, , /** and is useful for entering numerical data quickly.

## How a Keyboard Works — Step by Step

1. The user presses a key.
2. The keyboard's internal controller detects exactly which key was pressed and generates a unique scan code for that key.
3. This scan code is sent from the keyboard to the computer, usually over a USB connection.
4. The computer's keyboard driver translates the scan code into a character code (such as ASCII or Unicode), based on the current keyboard layout.
5. The resulting character is passed to the active application and displayed on screen.

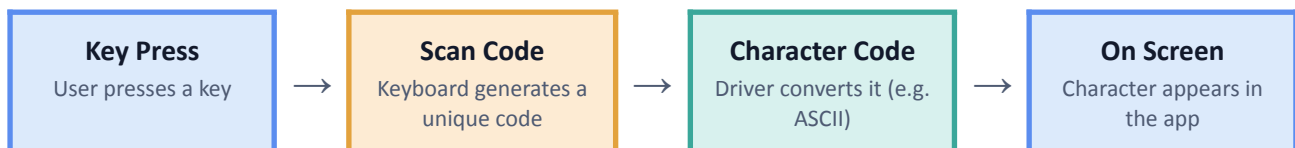


Fig 2.1 — How a key press becomes a character on screen.

### Important Point for Exams

A scan code and a character code are NOT the same thing. The scan code identifies WHICH physical key was pressed; the character code is the meaningful letter or symbol displayed, decided only after translation. This distinction is frequently tested.

## 4 The Mouse

The mouse is a pointing input device used to control the position of the on-screen cursor and to select or interact with items via its buttons. . It helps the user to select, open, move, and perform different operations on items displayed on the screen..

**The main parts of a mouse are:**

1. **Left Button:** It is used to select items, open files and folders, and perform different commands. A single click selects an item, while a double-click usually opens it.
2. **Right Button:** It is used to display a **shortcut or context menu** containing additional options related to the selected item.
3. **Scroll Wheel:** It is located between the left and right buttons. It is used to scroll a page or document **up and down**.
4. **Sensor:** An optical or laser sensor is present at the bottom of the mouse. It detects the movement of the mouse and moves the pointer accordingly on the screen.

**Common Mouse Operations:**

- **Pointing:** Moving the mouse to place the pointer on an item.
- **Clicking:** Pressing and releasing the left mouse button once.
- **Double-clicking:** Pressing the left button twice quickly, usually to open a file or folder.
- **Right-clicking:** Pressing the right mouse button to display a shortcut menu.

- **Dragging and Dropping:** Holding the left button while moving the mouse to move an item from one place to another.
- **Scrolling:** Rotating the scroll wheel to move up or down a page.

### How an Optical Mouse Works — Step by Step

1. A small LED on the underside of the mouse shines light onto the surface below it.
2. A tiny image sensor captures thousands of images of that surface every second.
3. A built-in processor compares each new image with the previous one to calculate the direction and distance moved.
4. This movement is converted into a change in x and y coordinates and sent to the computer.
5. The computer moves the on-screen cursor by exactly that amount.

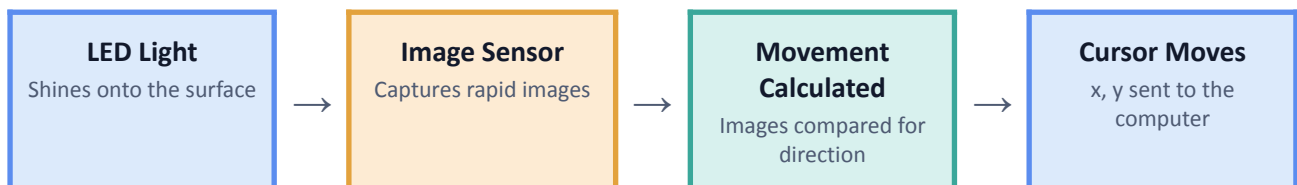


Fig 2.2 — How an optical mouse detects and reports movement.

**Older mechanical mice** used a rubber ball that rolled as the mouse moved, turning internal rollers to measure motion — now obsolete, replaced entirely by optical and laser sensors.

## 5 Touch Screens — Resistive & Capacitive

A touch screen is an input (and output) device that allows a user to interact directly with what is displayed, by touching the screen itself rather than using a separate device like a mouse. The two most widely used touch technologies work in fundamentally different ways.

### Working of a Touch Screen:

When a user touches the screen, the touch-sensitive surface detects the **position of the touch**. The device processes this information and performs the required action, such as opening an application, selecting an option, or typing a character.

### Common Touch Screen Operations:

1. **Tap:** Touch the screen once to select or open an item.
2. **Double Tap:** Tap twice quickly to perform certain actions.
3. **Swipe:** Move a finger across the screen to change pages or scroll.
4. **Drag:** Touch and move an item from one position to another.
5. **Pinch:** Move two fingers closer together to **zoom out**.
6. **Spread:** Move two fingers apart to **zoom in**.

## Uses of Touch Screens:

Touch screens are commonly used in **smartphones, tablets, ATMs, information kiosks, ticket machines, laptops, and self-service systems.**

| Touch Screen Technologies                       |                                                  |
|-------------------------------------------------|--------------------------------------------------|
| <b>Resistive</b><br>• Detects physical PRESSURE | <b>Capacitive</b><br>• Detects electrical CHARGE |

Fig 2.3 — The two main touch screen technologies.

## How a Resistive Touch Screen Works — Step by Step

1. Two thin, flexible, electrically conductive layers sit with a small gap between them.
2. Pressing the screen — with a finger, a stylus, or a gloved hand — pushes the top layer down to touch the bottom layer at that point.
3. This contact completes an electrical circuit at exactly that location.
4. The controller measures the resulting change in electrical resistance to calculate the precise x, y location touched.

## How a Capacitive Touch Screen Works — Step by Step

1. The screen's surface is coated with a material holding a uniform electrostatic charge.
2. A bare finger, being electrically conductive, draws away a tiny amount of this charge at the point of contact.
3. Sensors around the edges of the screen detect this localised change in capacitance.
4. The controller calculates the exact touch location from the pattern of change, and can track several touches at once, enabling multi-touch gestures like pinch-to-zoom.

| Point               | Resistive                                | Capacitive                                       |
|---------------------|------------------------------------------|--------------------------------------------------|
| What it detects     | Physical pressure between two layers     | Electrical charge from a conductive touch        |
| Works with          | Any object — finger, gloved hand, stylus | Bare finger, or a special conductive stylus only |
| Multi-touch support | Poor / usually not supported             | Excellent — supports gestures like pinch-to-zoom |
| Typical devices     | Older ATMs, industrial equipment         | Smartphones, tablets, modern laptops             |

## 6 The Joystick

A joystick is an input device with a stick that pivots on a base, used to control direction and movement — widely used in gaming, flight simulators, cranes, and powered wheelchairs.

### How a Joystick Works — Step by Step

1. The user tilts the joystick handle in a direction.
2. Sensors in the base — typically potentiometers or Hall-effect sensors — measure the angle of tilt along the X and Y axes (and sometimes a twist or Z axis).
3. This physical tilt is converted into an electrical signal proportional to the direction and degree of movement.
4. The signal is sent to the computer, which moves an on-screen object or control accordingly.

**Common uses:** video games, flight and driving simulators, cranes and industrial machinery, powered wheelchairs.

## 7 The Stylus — Passive & Active

A stylus is a pen-shaped input device used to point, draw, or write directly on a touch-sensitive screen.

| Point                | Passive Stylus                                             | Active (Digital) Stylus                                         |
|----------------------|------------------------------------------------------------|-----------------------------------------------------------------|
| Electronics          | None — acts as a simple conductive substitute for a finger | Contains its own electronics and battery/charge                 |
| Works on             | Capacitive touch screens only                              | Screens with a matching digitiser layer                         |
| Pressure sensitivity | No                                                         | Yes — pressing harder produces a thicker/darker line            |
| Palm rejection       | No                                                         | Yes — the screen ignores a resting palm while the pen is in use |
| Typical use          | Basic navigation and tapping                               | Precise drawing, note-taking, design work                       |

## 8 Scanners & Readers

A scanner is an input device that converts a physical document, image, or mark into digital data. In general, a light source illuminates the document, and a sensor array captures the reflected light,

converting it into digital data pixel by pixel. Four specialised types are commonly examined, each reading a different kind of information.

| Scanners & Readers                                                                            |                                                                                      |                                                                                               |                                                                                                        |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>OCR</b> <ul style="list-style-type: none"> <li>• Reads printed/handwritten TEXT</li> </ul> | <b>OMR</b> <ul style="list-style-type: none"> <li>• Reads filled-in MARKS</li> </ul> | <b>MICR</b> <ul style="list-style-type: none"> <li>• Reads MAGNETIC INK characters</li> </ul> | <b>Bar Code Reader</b> <ul style="list-style-type: none"> <li>• Reads printed LINE patterns</li> </ul> |

Fig 2.4 — Four specialised scanning technologies, each reading a different kind of mark.

## 8.1 OCR — Optical Character Recognition

**OCR stands for Optical Character Recognition.**

OCR is a technology used to **recognize printed or handwritten text from a scanned document or image and convert it into machine-readable/editable digital text.**

For example, suppose a printed book page is scanned into a computer. Normally, the computer initially treats the scanned page as an **image**. OCR software analyzes the image, identifies characters such as A, B, C, 1, 2, 3, etc., and converts them into text that can be searched, copied, or edited.

### Working of OCR

The basic working process is:

**Printed Document → Scanner/Camera → Image → OCR Processing → Recognized Text → Editable/Searchable Document**

The main steps are:

1. The document is scanned or photographed.
2. A digital image of the document is created.
3. OCR software analyzes the image.
4. It detects lines, words, and characters.
5. Characters are recognized and converted into machine-readable text.
6. The recognized text can be stored, searched, or edited.

### Applications of OCR

OCR is used for:

- Digitization of old books
- Converting printed documents into editable text
- Digitization of newspapers
- Libraries and archives

- Processing forms and documents
- Extracting text from scanned PDFs
- Recognizing text from photographs
- Office automation
- Data entry

### Advantages

- Reduces manual typing.
- Saves time.
- Helps digitize paper documents.
- Makes scanned documents searchable.
- Reduces data-entry effort.
- Useful for preserving old records.

### Limitation

OCR accuracy can decrease when:

- Image quality is poor.
- Text is blurred.
- Fonts are unusual.
- Handwriting is difficult to read.
- The document is damaged.

## 8.2 OMR — Optical Mark Reader

**OMR stands for Optical Mark Recognition.**

OMR is a technology used to **detect marks made at predefined positions on specially designed forms or answer sheets.**

It is commonly used for checking **multiple-choice examination answer sheets.**

For example:

**Which is an input device?**

- Monitor
- Keyboard
- Printer
- Speaker

The candidate fills the appropriate circle/bubble. An OMR system detects which position has been marked.

## **Working of OMR**

The basic process is:

**OMR Sheet → OMR Scanner/Reader → Detection of Marks → Processing → Result**

Steps:

1. A specially designed OMR sheet is provided.
2. The user marks the required bubbles or boxes.
3. The sheet is passed through an OMR scanner/reader.
4. The system detects marks at predefined locations.
5. The detected responses are compared with stored data or an answer key.
6. Results are generated automatically.

## **Applications of OMR**

OMR is commonly used for:

- Competitive examinations
- Entrance examinations
- Objective-type tests
- Surveys
- Feedback forms
- Questionnaires
- Voting or polling forms in some systems
- Attendance forms

## **Advantages**

- Very fast processing.
- Suitable for processing a large number of forms.
- Reduces manual checking.
- Useful for objective examinations.
- Provides consistent automated processing.
- Saves time.

## **Limitations**

- Usually requires a specially designed form.
- Incorrectly filled bubbles may cause recognition problems.
- Extra/unintended marks can create errors.

- It recognizes **marks**, not general written text.

### 8.3 MICR — Magnetic Ink Character Reader

**MICR stands for Magnetic Ink Character Recognition.**

MICR is a character-recognition technology traditionally used mainly in the **banking industry for processing cheques and similar documents.**

Characters are printed using **special magnetic ink/toner** in a standardized machine-readable format.

On Indian cheques, the MICR information is traditionally printed near the bottom of the cheque.

#### **Working of MICR**

The general process is:

**Cheque → MICR Reader → Magnetic Characters Read → Information Processed**

1. Special characters are printed using magnetic ink/toner.
2. The cheque is passed through compatible processing equipment.
3. The printed characters are magnetized and detected by the MICR reader.
4. The system recognizes the encoded information.
5. The information is used during cheque processing.

#### **Applications of MICR**

MICR has mainly been used in:

- Banks
- Cheque processing
- Automated cheque clearing
- Identification of bank/branch information
- High-volume financial document processing

#### **Advantages**

- Fast processing of cheques.
- High accuracy.
- Suitable for processing large numbers of banking documents.
- Characters can remain machine-readable even when ordinary visual marks or stamps are present, subject to document condition.

#### **Limitations**

- Requires specialized printing and reading technology.
- More expensive than ordinary printed text.
- Mainly used for specialized banking/document-processing purposes.

### Why MICR Is Used for Cheques

Magnetic ink is very difficult to forge or alter, and can still be read even if the cheque is stamped, signed over, or partially obscured — making MICR extremely reliable for banking.

## 8.4 Bar Code Reader

A bar code reader reads a printed pattern of parallel lines of varying widths and spacing, decoding it into a number that identifies a product or item.

1. A light source (LED or laser) shines across the barcode.
2. A sensor measures the light reflected back, which varies depending on whether it strikes a light space or a dark bar.
3. This pattern of light and dark is converted into an electrical signal.
4. The signal is decoded, based on the specific barcode standard used, into the numeric or alphanumeric code it represents.

**Uses:** retail checkout billing, inventory and warehouse tracking, library book checkouts.

### Scanners & Readers — Full Comparison

| Type            | Reads                       | Common Use                         |
|-----------------|-----------------------------|------------------------------------|
| OCR             | Printed or handwritten text | Digitising documents and books     |
| OMR             | Filled-in marks / bubbles   | Exam answer sheets, surveys        |
| MICR            | Magnetic ink characters     | Bank cheques                       |
| Bar Code Reader | Printed line patterns       | Retail billing, inventory tracking |

## 9 Biometric Devices

A **biometric device** scans and digitally records a person's unique physical or behavioural characteristics, used to verify their identity.

| Biometric Devices                                                                               |                                                                                            |                                                                                                    |                                                                                                 |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Fingerprint</b> <ul style="list-style-type: none"> <li>Ridge &amp; valley pattern</li> </ul> | <b>Iris / Retina</b> <ul style="list-style-type: none"> <li>Unique eye patterns</li> </ul> | <b>Facial Recognition</b> <ul style="list-style-type: none"> <li>Facial feature spacing</li> </ul> | <b>Voice Recognition</b> <ul style="list-style-type: none"> <li>Pitch, tone, cadence</li> </ul> |

Fig 2.5 — Common types of biometric device.

### How Fingerprint Recognition Works — Step by Step

1. The sensor (optical or capacitive) captures an image of the ridge-and-valley pattern on the fingertip.
2. Software analyses the image and extracts unique reference points, called minutiae, where ridges split, end, or merge.
3. This set of minutiae is converted into a digital template.
4. The new template is compared against templates stored during an earlier enrolment; if enough minutiae match closely enough, the fingerprint is accepted, otherwise it is rejected.

| Type                                                                                                                                                                                                                                                                | What It Scans                                                             | How                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Iris / Retina Scanner                                                                                                                                                                                                                                               | The coloured ring (iris) or blood vessels at the back of the eye (retina) | A camera captures the unique pattern, compared against a stored template           |
| Facial Recognition                                                                                                                                                                                                                                                  | The spacing and shape of facial features                                  | A camera maps the face into a digital template, compared against a stored template |
| Voice Recognition                                                                                                                                                                                                                                                   | Vocal characteristics such as pitch, tone, and cadence                    | A microphone captures speech, analysed and compared against a stored voiceprint    |
| <b>Real-Life Example</b><br>Unlocking a smartphone with a fingertip, an office using facial recognition instead of a swipe card, and India's Aadhaar system enrolling citizens using both fingerprint and iris scanning are all everyday uses of biometric devices. |                                                                           |                                                                                    |

## 10 Digital Cameras

A **digital camera** is an electronic device used to **capture photographs and videos in digital form**. Unlike traditional film cameras, a digital camera does not use photographic film. Instead, it uses an **electronic image sensor** to capture light and stores the resulting image as a digital file in internal memory or on a memory card.

Digital cameras are used in mobile phones, professional photography, scientific research, security systems, journalism, education, and many other fields.

### Basic Working of a Digital Camera

The working of a digital camera can be understood in the following steps:

#### 1. Light enters through the lens:

When the camera is pointed toward an object, light reflected from the object enters through the **camera lens**.

#### 2. Lens focuses the light:

The lens focuses the incoming light onto an electronic **image sensor**.

#### 3. Image sensor captures the image:

The image sensor converts incoming light into electrical signals. Two commonly used image-sensor technologies are:

- **CCD – Charge-Coupled Device**
- **CMOS – Complementary Metal-Oxide-Semiconductor**

#### 4. Image processing:

The camera's processor processes these electrical signals and converts them into a digital image.

#### 5. Image storage:

The final image is stored as a digital file, commonly in formats such as **JPEG** or **RAW**, in internal storage or on a memory card.

### Simple Flow

**Object → Light → Lens → Image Sensor → Image Processor → Digital Image → Storage**



Fig 2.6 — How a digital camera captures and stores a photo.

Digital cameras provide several advantages:

1. **Immediate Preview:** Photographs can be viewed immediately after capturing them.
2. **No Film Required:** Images are stored electronically.
3. **Large Storage Capacity:** Hundreds or thousands of photographs can be stored depending on capacity and file size.
4. **Easy Deletion:** Unwanted photographs can be deleted.
5. **Easy Transfer:** Images can be transferred to computers and other devices.
6. **Editing:** Digital photographs can easily be edited using image-editing software.
7. **Video Recording:** Most modern digital cameras can also record videos.
8. **Easy Sharing:** Digital images can easily be shared through the Internet.
9. **Reusable Storage:** Memory cards can normally be erased and reused.
10. **Better Organization:** Photographs can be organized into folders and albums.

Some disadvantages are:

- They require battery power.
- High-quality professional cameras can be expensive.
- Memory cards can become corrupted or damaged.
- Digital files may be accidentally deleted.
- Large high-resolution photographs require more storage space.
- Cameras and sensors can be damaged by water, dust, or physical impact.
- Learning advanced camera settings can require practice.

## 11 Full Recap — All Input Devices

| Device                  | Converts                                      | Key Mechanism                                      |
|-------------------------|-----------------------------------------------|----------------------------------------------------|
| Keyboard                | Key press → character                         | Scan code, translated into a character code        |
| Mouse                   | Hand movement → cursor movement               | LED + image sensor detects motion                  |
| Resistive Touch Screen  | Pressure → location                           | Two layers touch, resistance change measured       |
| Capacitive Touch Screen | Conductive touch → location                   | Electrostatic field disrupted                      |
| Joystick                | Stick tilt → direction signal                 | Potentiometers / Hall-effect sensors measure angle |
| Stylus                  | Pen contact → position (+ pressure if active) | Capacitive substitute, or digitiser communication  |
| OCR                     | Printed/handwritten text → editable text      | Character shape pattern matching                   |
| OMR                     | Filled marks → data                           | Reflected light measured at mark positions         |
| MICR                    | Magnetic ink characters → data                | Magnetic waveform read                             |
| Bar Code Reader         | Printed lines → ID code                       | Reflected light pattern decoded                    |
| Biometric (Fingerprint) | Physical trait → identity match               | Minutiae comparison against a stored template      |
| Digital Camera          | Light → digital image                         | Image sensor converts light to electrical charge   |

## 12 Putting It All Together — A Worked Example

Consider a university conducting a semester examination for 500 students:

| Step | Device Used                     | What Happens                                                                                                                             |
|------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Biometric (Fingerprint) Scanner | Each student's identity is verified at the door using a fingerprint scanner before entry to the exam hall.                               |
| 2    | Bar Code Reader                 | The student's admit card barcode is scanned to log their attendance against their exam registration.                                     |
| 3    | OMR Answer Sheet                | Students mark their objective-type answers by filling in bubbles on a printed OMR sheet.                                                 |
| 4    | OMR Scanner                     | After the exam, every answer sheet is passed through an OMR scanner, which reads the filled bubbles and calculates scores automatically. |
| 5    | Digital Camera                  | A digital camera captures each student's photo for the exam centre's attendance record.                                                  |
| 6    | Keyboard & Mouse                | Exam staff use a keyboard and mouse to enter and verify the results in the university's computer system.                                 |

This single real-world scenario uses six different input devices from this lecture, each solving a different part of the same overall task — verifying identity, recording attendance, capturing answers, and entering results.

## 13 Fill in the Blanks

- 1. The code generated when a key is pressed on a keyboard is called a(n) \_\_\_\_\_.
- 2. A modern optical mouse uses a(n) \_\_\_\_\_ and an image sensor to detect movement.
- 3. A \_\_\_\_\_ touch screen detects physical pressure, while a \_\_\_\_\_ touch screen detects electrical charge.
- 4. A joystick uses \_\_\_\_\_ or Hall-effect sensors to measure the angle of tilt.
- 5. A(n) \_\_\_\_\_ stylus contains its own electronics and can detect pressure, unlike a passive stylus.
- 6. \_\_\_\_\_ technology converts printed or handwritten text in a scanned image into editable text.
- 7. \_\_\_\_\_ detects the presence of filled-in marks on a form, commonly used for exam answer sheets.
- 8. \_\_\_\_\_ reads characters printed using magnetised ink, most commonly found on bank cheques.
- 9. In fingerprint recognition, the unique reference points where ridges split or end are called \_\_\_\_\_.
- 10. A digital camera's image sensor is commonly either a CCD or a(n) \_\_\_\_\_ chip.

### Answers

1. Scan code | 2. LED | 3. Resistive, Capacitive | 4. Potentiometers | 5. Active (digital) | 6. OCR (Optical Character Recognition) | 7. OMR (Optical Mark Reader) | 8. MICR (Magnetic Ink Character Reader) | 9. Minutiae | 10. CMOS

## 14 True or False

- 1. A passive stylus can detect how hard the user is pressing on the screen.
- 2. OMR is commonly used to check multiple-choice examination answer sheets.
- 3. MICR ink is difficult to forge, making it useful for printing bank cheques.
- 4. A capacitive touch screen can be reliably operated using a plain, non-conductive stylus.
- 5. A joystick's tilt is converted into an electrical signal by sensors in its base.
- 6. OCR technology detects filled-in bubbles rather than recognising text.
- 7. Fingerprint recognition compares captured minutiae points against a stored template.
- 8. A digital camera's image sensor converts light into an electrical charge.

### Answers with Explanation

- 1. False.** A passive stylus has no electronics and cannot detect pressure; that is a feature of an active (digital) stylus.
- 2. True.** OMR is precisely designed for this task.
- 3. True.** Magnetic ink is very hard to forge or erase, making MICR highly secure and reliable for banking.
- 4. False.** A capacitive screen needs a conductive touch; a plain, non-conductive stylus generally will not register.
- 5. True.** Potentiometers or Hall-effect sensors in the base measure the tilt and convert it into an electrical signal.
- 6. False.** That describes OMR; OCR recognises and converts text characters, not filled marks.
- 7. True.** That is exactly how fingerprint matching works — comparing extracted minutiae against a stored template.
- 8. True.** That is the core function of an image sensor, whether CCD or CMOS.

## 15 Short Answer Questions (2–3 Marks)

- Differentiate between a scan code and a character code. (2 marks)
- Explain, in brief, how an optical mouse detects movement. (3 marks)
- Differentiate between a resistive and a capacitive touch screen. (3 marks)
- What is a joystick used for? Name two applications. (2 marks)
- Differentiate between a passive stylus and an active stylus. (3 marks)
- What is OCR? Give one use. (2 marks)
- Differentiate between OMR and MICR. (3 marks)
- How does a barcode reader identify a product? (2 marks)
- What is a biometric device? Name any two types. (2 marks)
- Explain, in brief, how a digital camera captures an image. (3 marks)

## 16 Long Answer Model Answers

The following are fully written model answers. Use these as a template for structure, depth, and the level of explanation expected in your own exam answer — do not simply memorise them word for word; practise writing them in your own words using the same structure.

### Q1 (5 Marks): Explain, step by step, how a keyboard and a mouse each convert a physical action into computer input.

A keyboard converts a key press through a two-step translation. First, when a key is pressed, the keyboard's internal controller generates a unique scan code identifying exactly which key was used. Second, the computer's keyboard

driver translates that scan code into a meaningful character code, such as ASCII, and this character is displayed in the active application.

A mouse converts hand movement using light and an image sensor. An LED on its underside shines onto the surface below, and a tiny image sensor captures thousands of images of that surface every second. A built-in processor compares consecutive images to calculate the direction and distance moved, converts this into a change in x and y coordinates, and sends it to the computer, which moves the on-screen cursor by that same amount.

**Q2 (10 Marks): Differentiate between resistive and capacitive touch screens, explaining how each detects a touch, with examples.**

| Touch Screen Technologies                       |                                                  |
|-------------------------------------------------|--------------------------------------------------|
| <b>Resistive</b><br>• Detects physical PRESSURE | <b>Capacitive</b><br>• Detects electrical CHARGE |

*(The two touch screen technologies, as in Fig 2.3 of these notes.)*

A resistive touch screen is built from two thin, flexible, conductive layers separated by a small gap. Pressing anywhere on the screen — with a finger, a stylus, or even a gloved hand — pushes the two layers together at that point, completing an electrical circuit; the controller calculates the touch location from the resulting change in electrical resistance. Because it responds to pressure from any object, a resistive screen works even with a gloved hand, but it generally cannot support true multi-touch gestures.

A capacitive touch screen is coated with a material holding a uniform electrical charge across its surface. A bare human finger, being electrically conductive, draws away a tiny amount of this charge at the point of contact; sensors around the screen detect precisely where this change in capacitance occurred. This method depends on conductivity rather than pressure, so a capacitive screen does not respond to a plain gloved hand, but it is far more sensitive, supporting multi-touch gestures such as pinch-to-zoom. Older bank ATMs commonly use resistive screens, while smartphones use capacitive screens.

**Q3 (10 Marks): Explain OCR, OMR, MICR, and Bar Code readers, describing what each reads, step by step, and a typical use for each.**

| Scanners & Readers                             |                                       |                                                |                                                         |
|------------------------------------------------|---------------------------------------|------------------------------------------------|---------------------------------------------------------|
| <b>OCR</b><br>• Reads printed/handwritten TEXT | <b>OMR</b><br>• Reads filled-in MARKS | <b>MICR</b><br>• Reads MAGNETIC INK characters | <b>Bar Code Reader</b><br>• Reads printed LINE patterns |

*(The four scanning technologies, as in Fig 2.4 of these notes.)*

| Type            | How It Works (Step by Step, Summarised)                                                                                       | Use                            |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| OCR             | Scans the document, segments the image into characters, matches each shape against known patterns, and outputs editable text. | Digitising books and documents |
| OMR             | Shines light onto a form, measures reflected light at each predefined position, and records low-reflection spots as marked.   | Exam answer sheets, surveys    |
| MICR            | Magnetises the document, passes it under a read head, and identifies each character from its unique magnetic waveform.        | Bank cheques                   |
| Bar Code Reader | Shines light across the barcode, measures the varying reflection from bars and spaces, and decodes the resulting pattern.     | Retail billing, inventory      |

In conclusion, all four technologies follow the same broad pattern — shine light or apply a signal, measure the response, and decode it — but each is tuned to read a completely different kind of mark, which is exactly why each has become the standard tool for its specific real-world job.

**Q4 (10 Marks): Explain what a biometric device is, and describe, step by step, how fingerprint recognition works. Briefly describe three other biometric methods.**

A biometric device scans and digitally records a person's unique physical or behavioural characteristics, used to verify their identity, rather than relying on something the person carries (like a card) or remembers (like a password).

1. The sensor (optical or capacitive) captures an image of the ridge-and-valley pattern on the fingertip.
2. Software analyses the image and extracts unique reference points, called minutiae, where ridges split, end, or merge.
3. This set of minutiae is converted into a digital template.
4. The new template is compared against templates stored during an earlier enrolment; a close enough match is accepted, otherwise it is rejected.

| Type                  | What It Scans                                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Iris / Retina Scanner | The coloured ring of the eye (iris) or the blood vessels at the back of the eye (retina), captured by camera and compared to a stored template. |
| Facial Recognition    | The spacing and shape of facial features, mapped by camera into a digital template and compared to a stored template.                           |
| Voice Recognition     | Vocal characteristics such as pitch, tone, and cadence, captured by microphone and compared to a stored voiceprint.                             |

In conclusion, every biometric method follows the same underlying logic as fingerprint recognition — capture a unique characteristic, convert it into a digital template, and compare it against a template stored earlier — differing only in which physical or behavioural trait is measured.

**Q5 (15 Marks): Describe, with a suitable example, how a university examination centre could use multiple input devices covered in this lecture together, explaining the role of each.**

Consider a university conducting a semester examination for 500 students, which draws on almost every device studied in this lecture, each solving a different part of the same overall task.

| Step | Device Used                     | Role                                                                                                             |
|------|---------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1    | Biometric (Fingerprint) Scanner | Verifies each student's identity at the door, comparing a live scan against their enrolled fingerprint template. |
| 2    | Bar Code Reader                 | Scans the student's admit card barcode to log their attendance against their exam registration.                  |
| 3    | OMR Answer Sheet                | Students record their objective-type answers by filling in bubbles on a printed sheet.                           |
| 4    | OMR Scanner                     | Reads every filled bubble after the exam and calculates scores automatically, far faster than manual checking.   |
| 5    | Digital Camera                  | Captures each student's photo for the exam centre's attendance and identity record.                              |
| 6    | Keyboard & Mouse                | Used by exam staff to enter and verify the final results in the university's computer system.                    |

Each device is chosen specifically for the job it does best: the fingerprint scanner solves the problem of confirming identity without relying on an easily forged ID card; the barcode reader makes attendance logging fast and error-free compared to manual entry; the OMR system allows hundreds of answer sheets to be scored in minutes rather than hours; and the digital camera, keyboard, and mouse handle record-keeping and result entry.

In conclusion, no single input device could manage this entire process alone; it is the combination of devices — each converting a different kind of real-world action or object into digital data — that allows a task as large as a university-wide examination to run smoothly, securely, and efficiently.

## 17 Exam Tips

- Memory trick for the keyboard: “Press → Scan code → Character code → Screen.”
- Memory trick for touch screens: Resistive = PRESSURE (works with anything); Capacitive = CHARGE (needs a bare finger or special stylus).
- Memory trick for scanners: OCR reads TEXT, OMR reads MARKS, MICR reads MAGNETIC INK, Bar Code readers read LINES.
- For “explain the working of...” questions (keyboard, mouse, touch screen, joystick, OCR/OMR/MICR/bar code, fingerprint, camera), always answer as a NUMBERED, step-by-step sequence — exactly like the model answers in Section 16.
- For “differentiate” questions (resistive vs capacitive, passive vs active stylus, OMR vs MICR), always answer using a table — it is faster to write and easier for the examiner to award marks clearly.
- Always draw the relevant diagram (Fig 2.1–2.6) whenever a question asks you to “explain” or “describe” a device's working — diagrams carry dedicated marks in most university marking schemes.
- Common Mistake: Do not confuse OCR (reads text) with OMR (reads filled-in marks) — they sound similar but do very different jobs.
- Common Mistake: Do not say a passive stylus can sense pressure — only an active (digital) stylus can, because only it contains its own electronics.
- For 10 and 15 mark questions, structure your answer with a clear paragraph or table per device, exactly like the model answers in Section 16 — examiners reward clear structure over long, unstructured paragraphs.

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*MDC Section II, Lecture 2 — Input Devices — how the real world gets converted into the data Lecture 1's memory and storage systems hold.*